Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title	**ELE827**
Course Name	**CMOS Mixed Signal**
Semester/Year	**F2026**

Instructor	**Dr Fei Yuan**

Lab Number	**Lab 5**
Lab Name	**SAR ADC**

Submission Date	
Due Date	

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab reports your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

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List of Abbreviations

ADE — Analog Design Environment
ADE XL — Analog Design Environment Explorer
TB — Testbench
TRAN — Transient Analysis
DC — Direct Current (DC analysis / operating point)
PVT — Process / Voltage / Temperature
TT / FF / SS — Typical / Fast / Slow process corners
VDD — Supply Voltage
GND — Ground
VIN — Input Voltage
VOUT — Output Voltage

Schematic

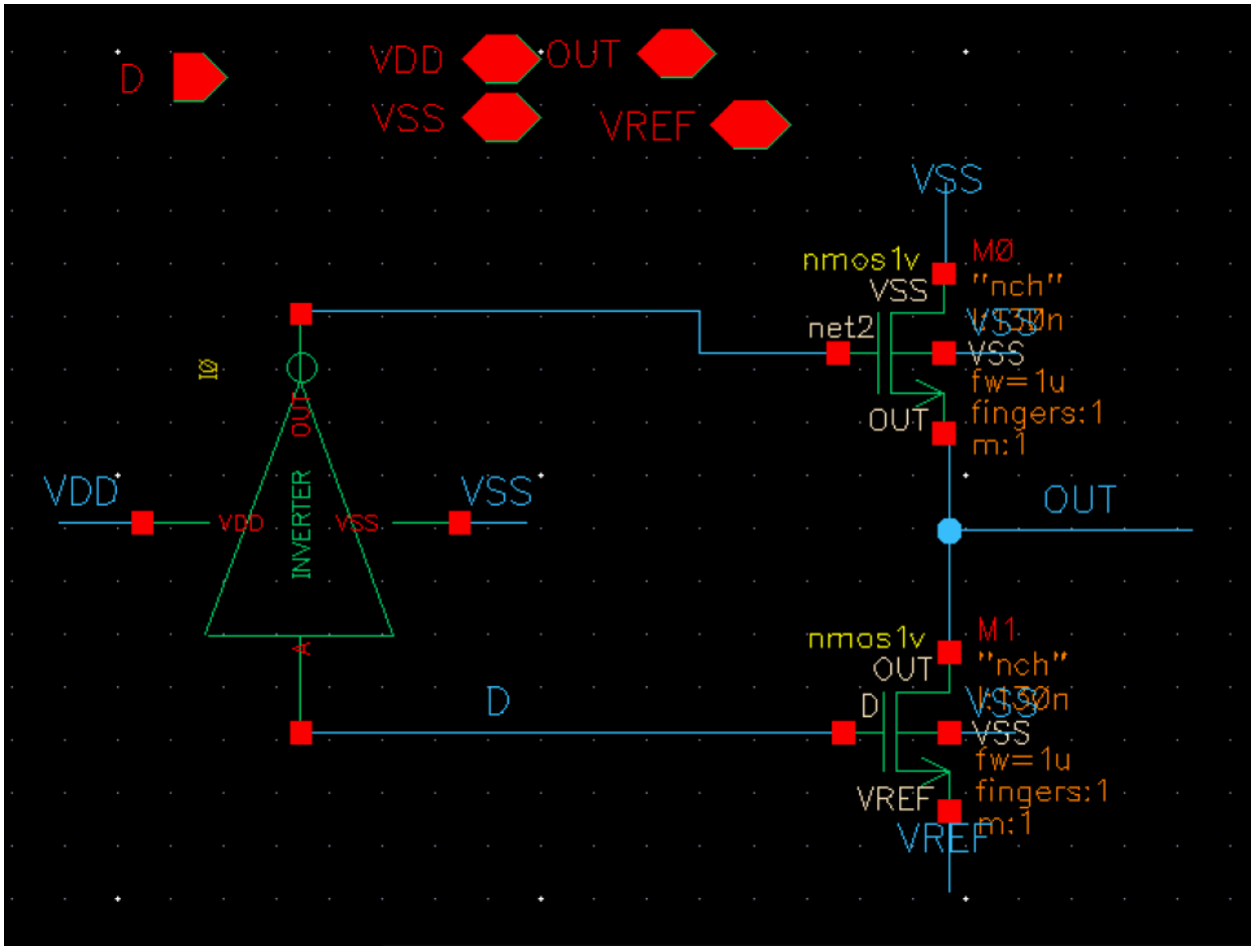


Figure 1: DAC Switches

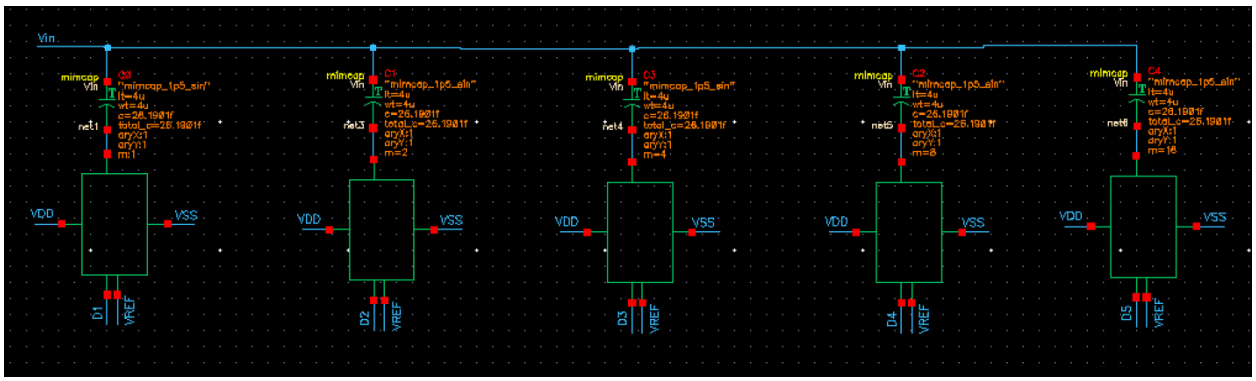


Figure 2: DAC

Component	Value
mimcap base	26.19 fF
Switch NMOS	300n

Table 1: Component Values for DAC

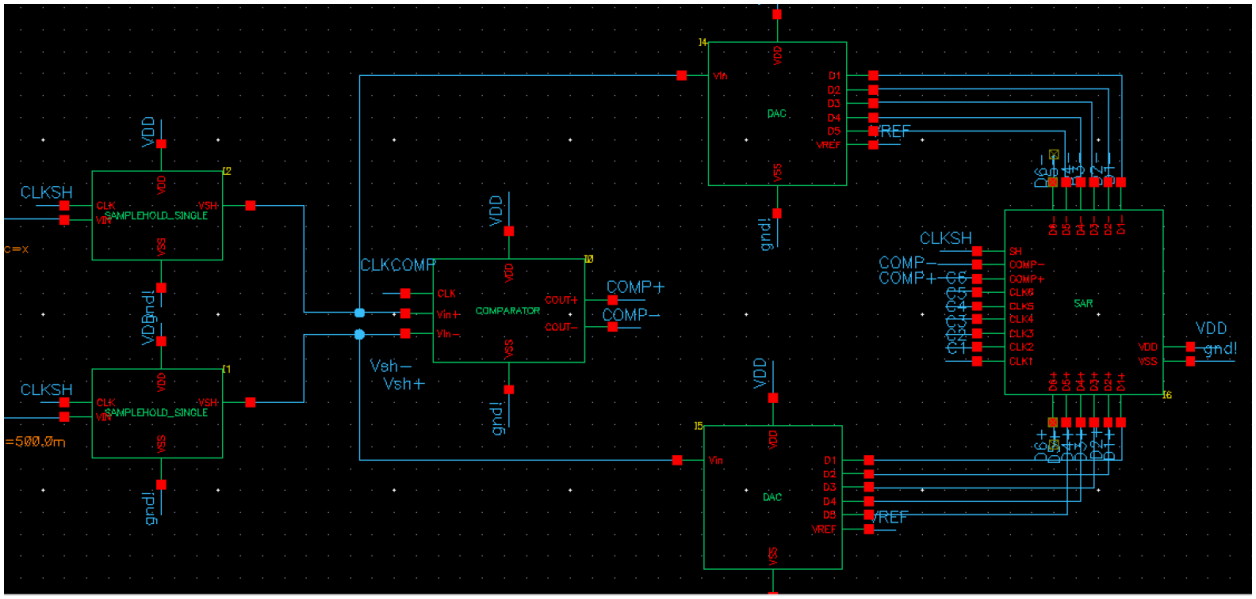


Figure 3: SAR ADC

DC Analysis

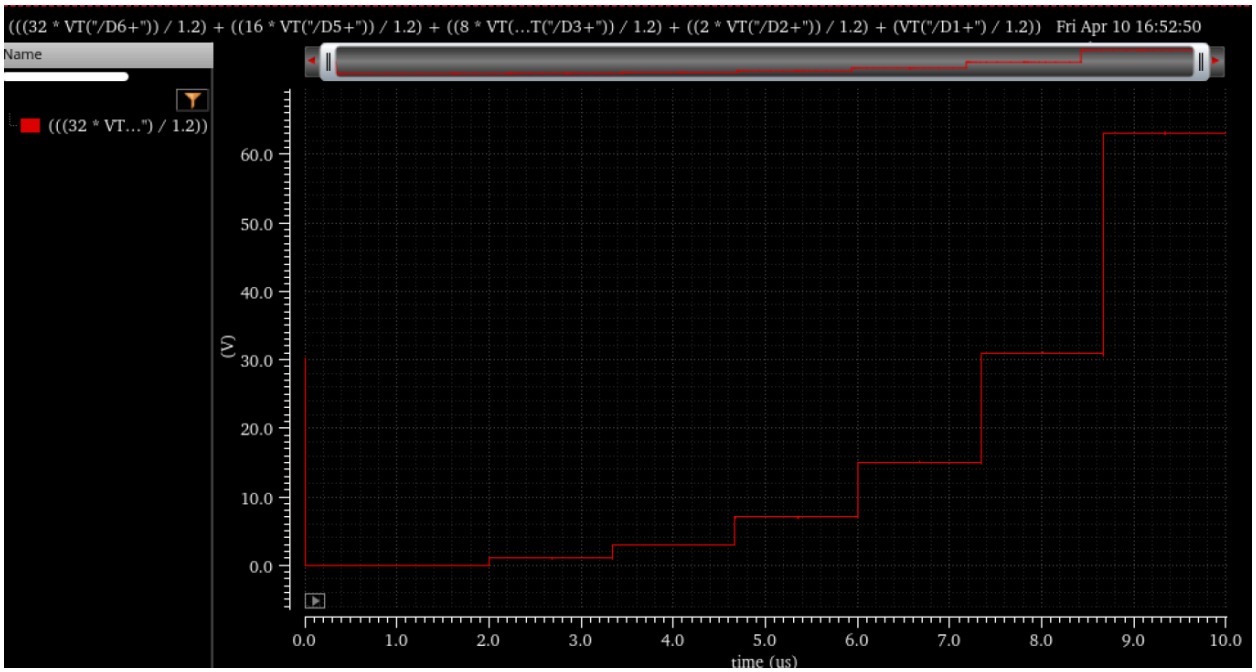


Figure 4: Sample Plot of DOUT+ with ΔV = .4

ΔV (V)	Dout (ideal)	Dout (simulation)	DNL	DIFF = Sim - Ideal	INL
0.0	0	42	—	42.0	42.0
0.1	16	47	5.0	31.0	73.0

0.2	32	31	-16.0	-1.0	72.0
0.3	48	31	0.0	-17.0	55.0
0.4	63	63.2	32.2	0.2	55.2

Table 2: DOUT Sweep Results

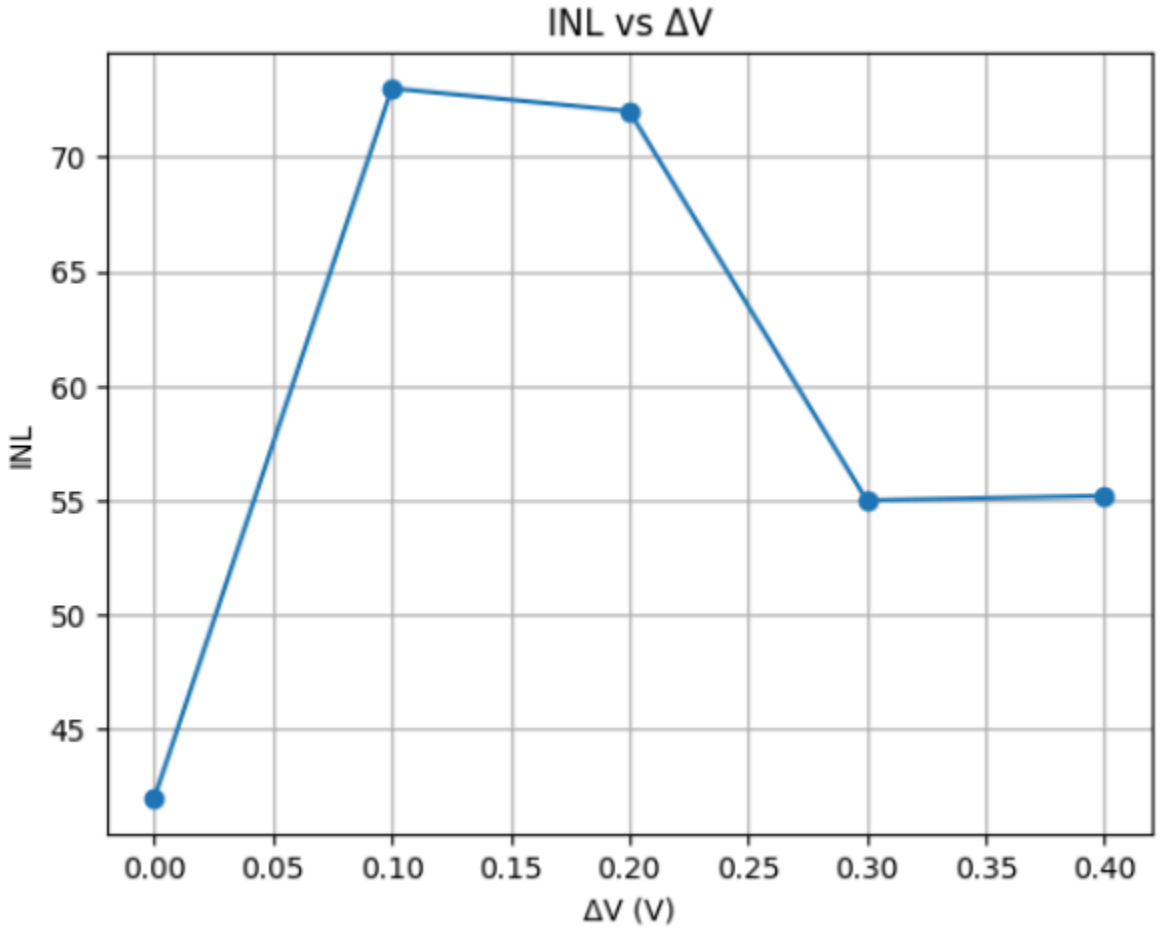


Figure 5: INL Plot

AC Analysis

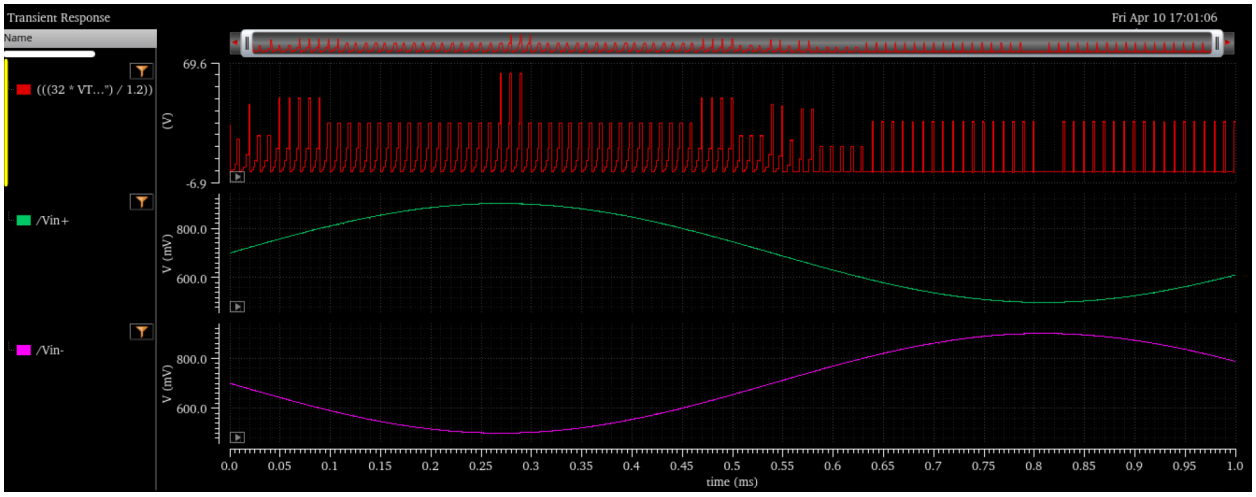


Figure 6: Sinusoidal input plot of DOUT+

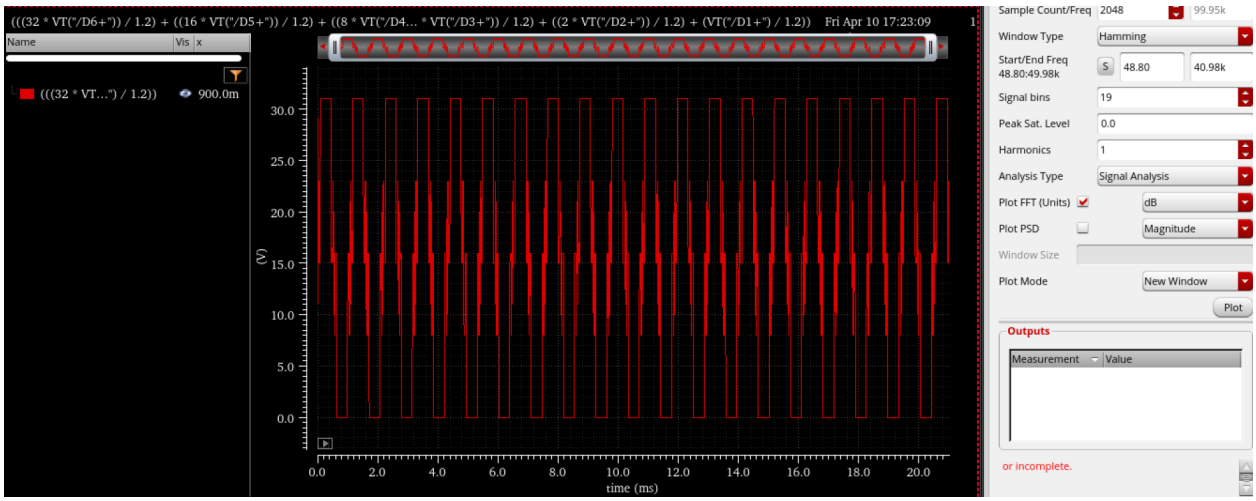


Figure 7: FFT Error Window

Conclusion

In conclusion, this lab uses all previous lab components such as the SAR, sample and hold, comparator, and DAC to form the SAR ADC. The sample and hold captures the input voltage during the sampling phase and holds it constant throughout the conversion process. The SAR then performs a binary sweep by setting each bit sequentially until it reaches its target. The DAC to generate a reference voltage that is compared against the held input using the comparator. Based on the comparator output, each bit is either kept or cleared, slowly refining the approximation. This process continues from the MSB to the LSB until the digital output is obtained. The SAR ADC in this case had several non-linearities which can be seen in **Table 1**, with the DOUT not following the ideal mapping causing it to go up and down when sweeping the differential voltage.